Project Design Phase
Solution Architecture

Date	15 February 2025
Team ID	LTVIP2025TMID45199
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

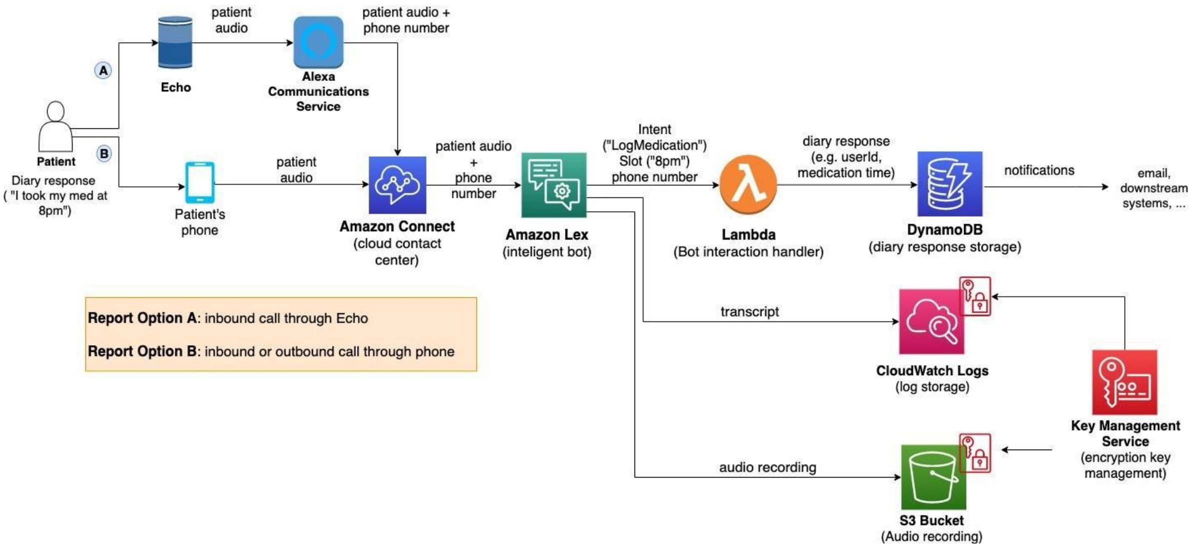

Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/

🔍 Solution Architecture Overview

Solution Architecture bridges the gap between biodiversity-related challenges and cutting-edge deep learning solutions. It defines the overall structure and flow of how data, models, and users interact with the application.

The architecture ensures real-time butterfly classification using pre-trained CNN models, while also considering scalability, modularity, and end-user usability.

🎯 Objectives of the Architecture

- **Identify butterfly species using image input from end users.**
- **Use transfer learning (VGG16) for efficient, high-accuracy prediction.**
- **Enable interactive and educational web-based classification interface.**
- **Ensure modular design for retraining, scaling to new species, or platform deployment.**

🧱 Solution Components

1. **Data Pipeline**
    - **Input: 6499 labeled images (75 butterfly species)**
    - **Image augmentation and preprocessing using Keras ImageDataGenerator**
2. **Model Architecture**
    - **Base Model: Pre-trained VGG16 (ImageNet weights)**
    - **Custom Layers: Flatten → Dense(256) → Dropout → Dense(75 softmax)**
3. **Training Environment**
    - **TensorFlow/Keras in Jupyter Notebook/Google Colab**
    - **GPU acceleration for training and fine-tuning**
4. **Model Evaluation**
    - **Accuracy and loss plots, confusion matrix**
    - **Final model exported (`.h5` format)**
5. **Deployment Layer**

- Web interface using Streamlit
- User uploads butterfly image → Receives predicted species with confidence score

6. **User Feedback Loop (optional future phase)**
   - Feedback collection for wrong predictions
   - Option to submit images to improve model accuracy over time

📊 **Architecture Diagram**